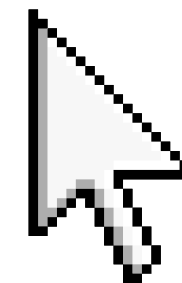


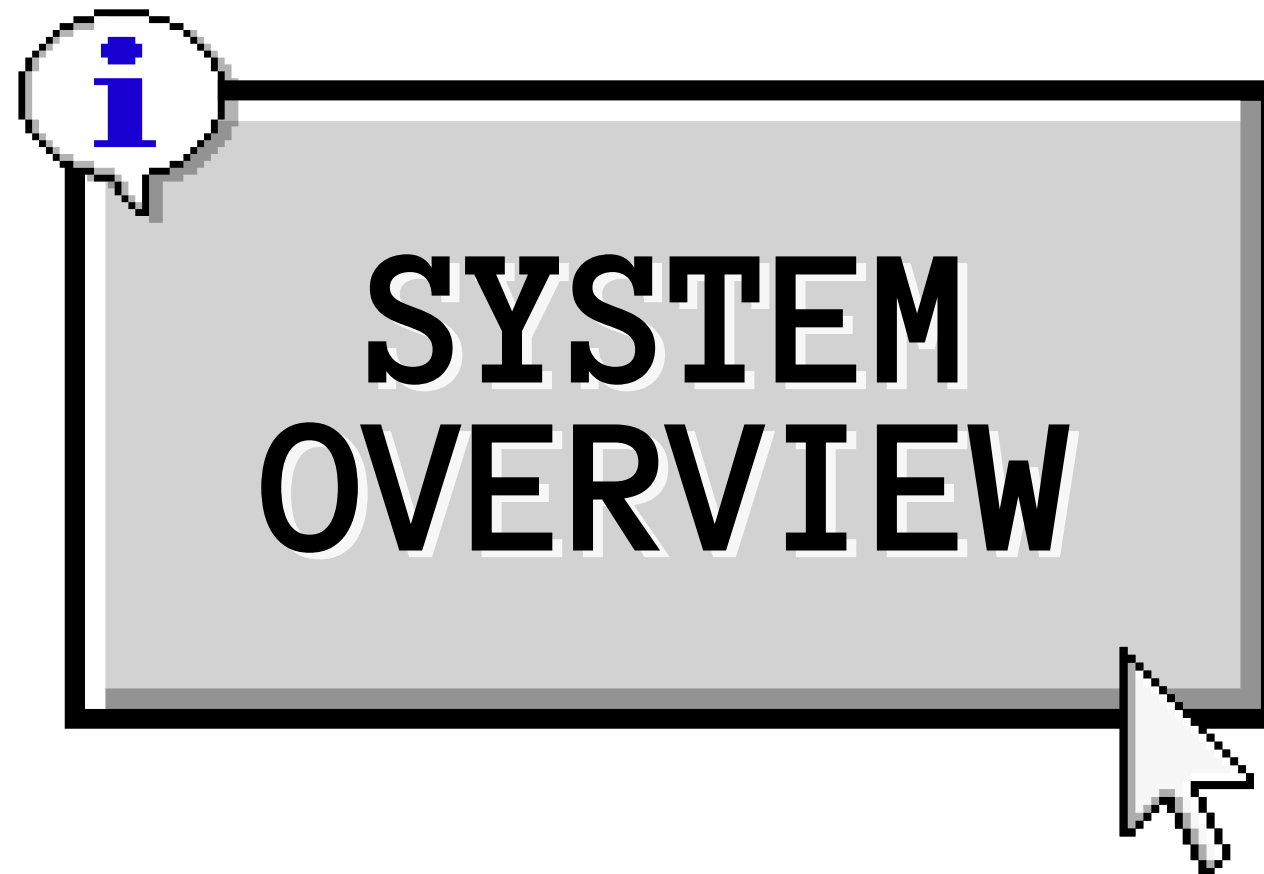
DBMS - 2

AIRPORT

ENG : AMMAR YASSER

Start





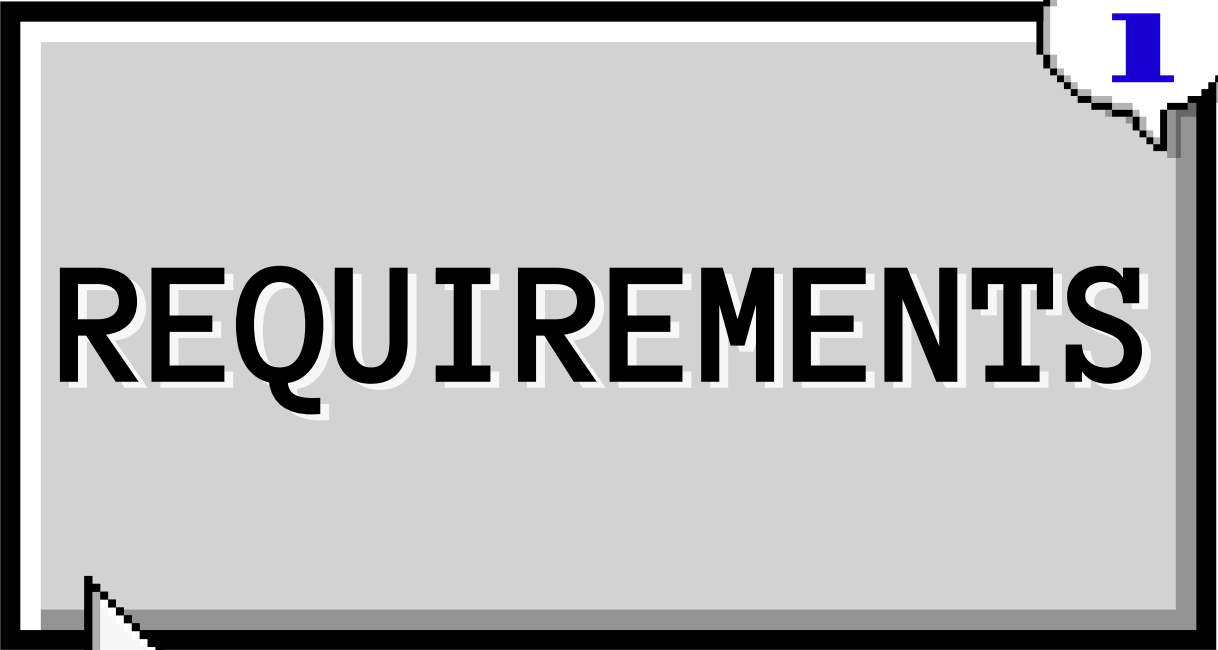
The Airport Management System is a database system designed to manage airport operations efficiently.

The system helps organize and store information related to airports, flights, passengers, bookings, tickets, staff, baggage, passports, and security checks.

The system aims to improve data management, reduce manual work, and provide fast access to airport information.

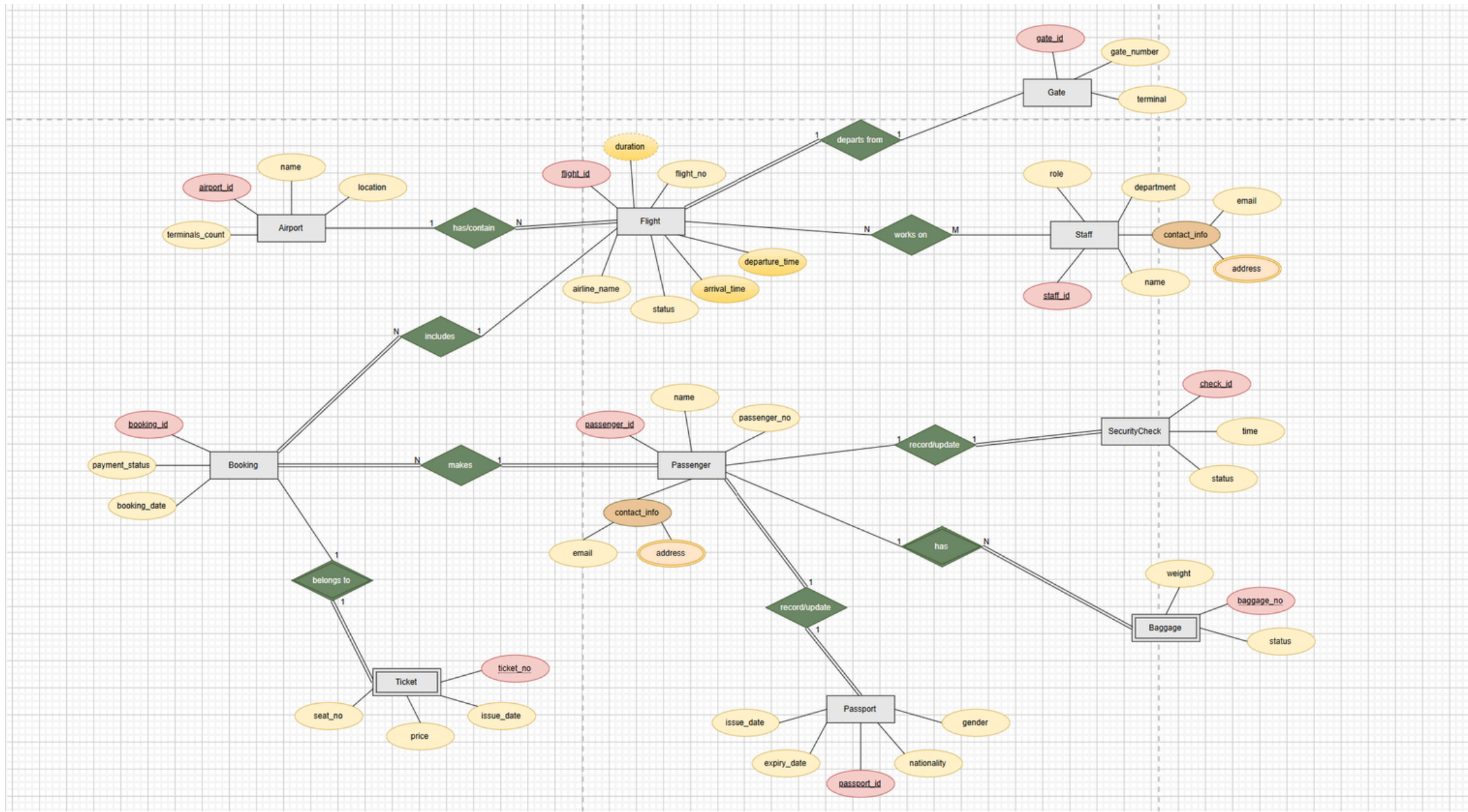
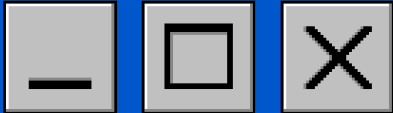


- Flights (numbers, times, status...)
- Gates & Airports
- Passengers & Passports
- Bookings, Tickets
- Security checks
- Staff working on flights
- Baggage per passenger



REQUIREMENTS

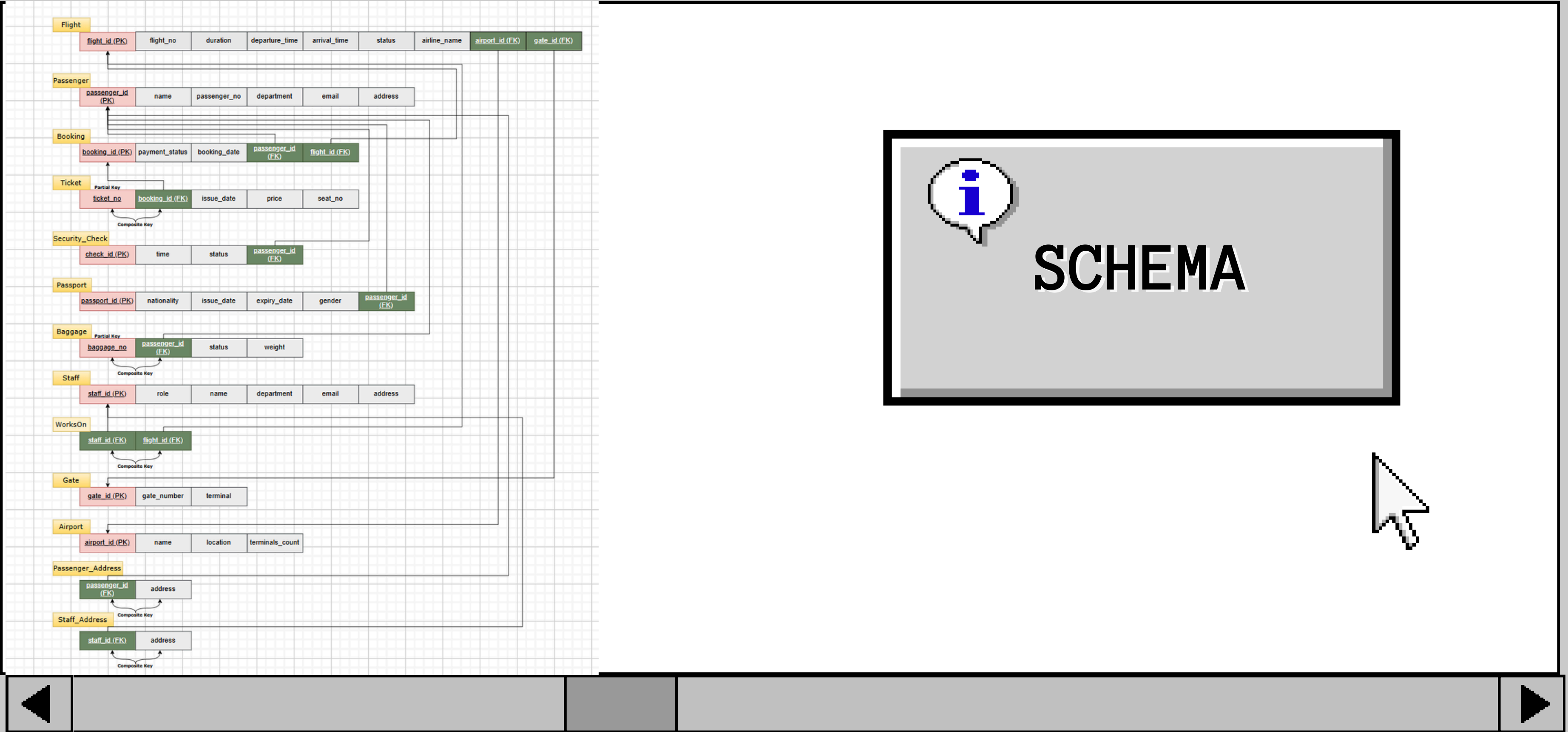






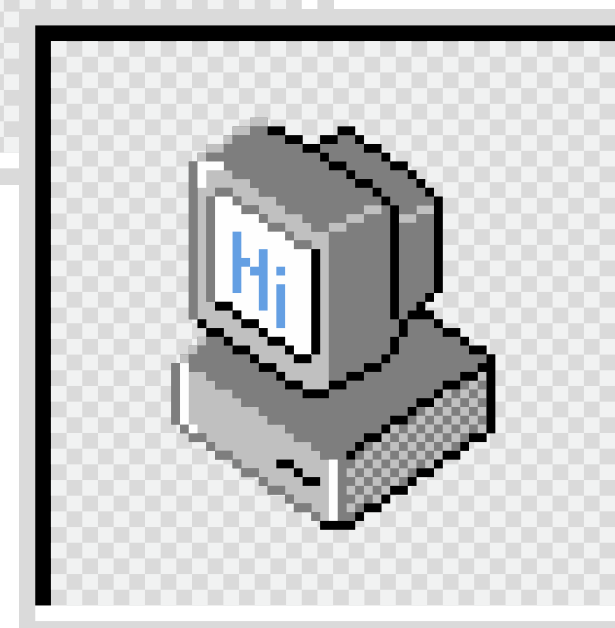
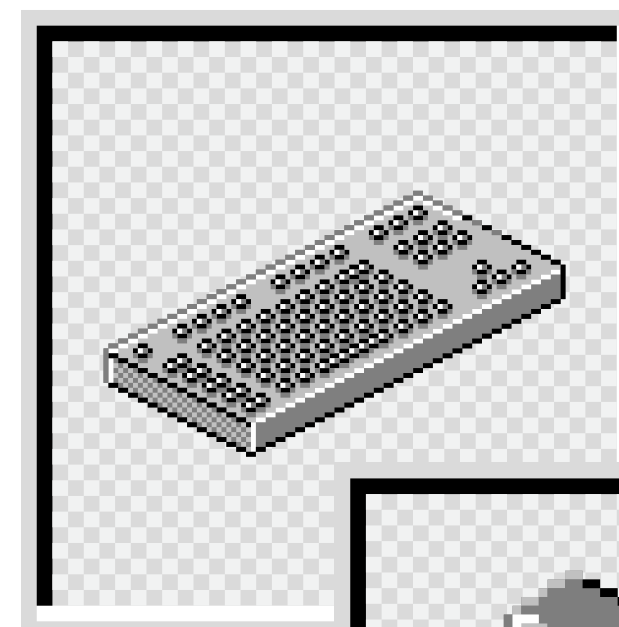
ENTITY
RELATIONSHIP
DIAGRAM

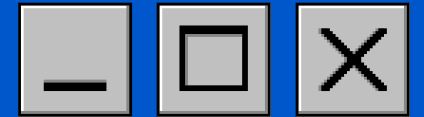






CREATING DATABASE





AIRPORT

Recovered-D...Alabd (88)* ✨ ✕ r5glbffy..sql...6\Alab

```
1
2 create database FINAL_AIRPORT;
3 USE FINAL_AIRPORT
4
5 create table airport (
6     airportId int primary key,
7     name varchar(100) not null,
8     location varchar(100) not null,
9     terminalsCount int not null
10 );
11
12 insert into airport values
13 (1, 'Cairo Airport', 'Cairo', 3),
14 (2, 'Alex Airport', 'Alexandria', 2),
15 (3, 'Aswan Airport', 'Aswan', 1),
16 (4, 'Luxor Airport', 'Luxor', 2),
17 (5, 'Hurghada Airport', 'Hurghada', 3);
18
```

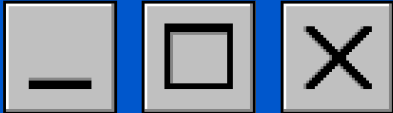


GATE

Recovered-D...Alabd (88)* ✨ ✕ r5glbffy..sql...6\Alabd (55))

```
19
20 create table gate (
21     gateId int primary key,
22     gateNumber varchar(20) not null,
23     terminal varchar(20) not null
24 );
25
26 insert into gate values
27 (1, 'G1', 'T1'),
28 (2, 'G2', 'T1'),
29 (3, 'G3', 'T2'),
30 (4, 'G4', 'T2'),
31 (5, 'G5', 'T3');
32
```

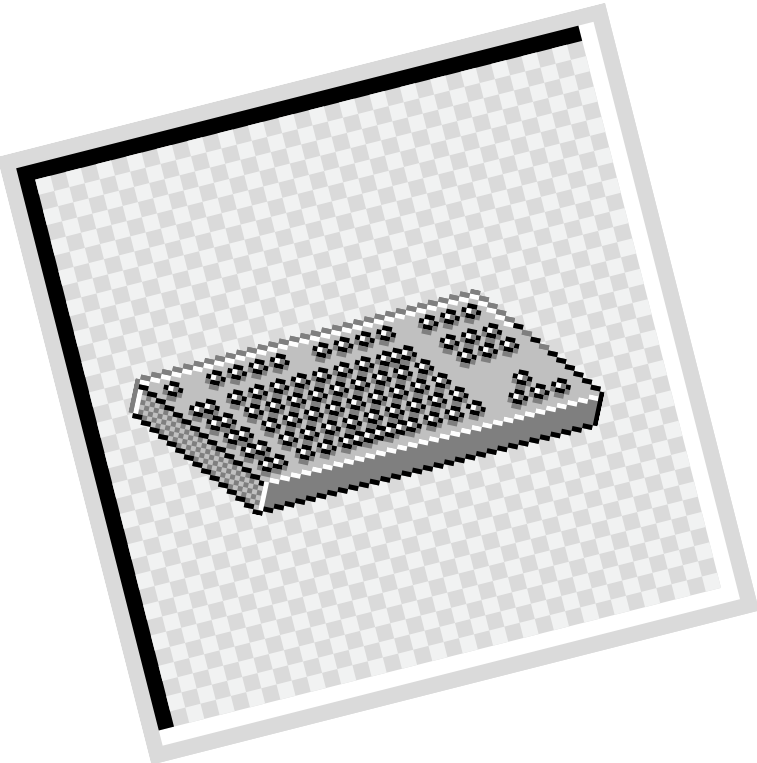


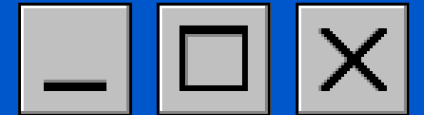


 FLIGHT

Recovered-D...Alabd (88))* r5glbffy..sql...6\Alabd (55))

```
34 |
35 | create table flight (
36 |     flightId int primary key,
37 |     flightNo varchar(50) not null,
38 |     duration varchar(50),
39 |     departureTime datetime not null,
40 |     arrivalTime datetime not null,
41 |     status varchar(50),
42 |     airlineName varchar(100) not null,
43 |     airportId int not null,
44 |     gateId int not null,
45 |     foreign key (airportId) references airport(airportId)
46 |         on delete cascade
47 |         on update cascade,
48 |     foreign key (gateId) references gate(gateId)
49 |         on delete no action
50 |         on update cascade
51 | );
52 |
53 | insert into flight values
54 | (1, 'MS101', '2h', '2025-01-01 08:00', '2025-01-01 10:00', 'on time', 'EgyptAir', 1, 1),
55 | (2, 'MS102', '1.5h', '2025-01-02 09:00', '2025-01-02 10:30', 'delayed', 'EgyptAir', 2, 2),
56 | (3, 'QR201', '3h', '2025-01-03 11:00', '2025-01-03 14:00', 'on time', 'Qatar Airways', 3, 3),
57 | (4, 'EK301', '4h', '2025-01-04 13:00', '2025-01-04 17:00', 'cancelled', 'Emirates', 4, 4),
58 | (5, 'SV401', '2.5h', '2025-01-05 15:00', '2025-01-05 17:30', 'on time', 'Saudi Airlines', 5, 5);
59 |
```





PASSENGER

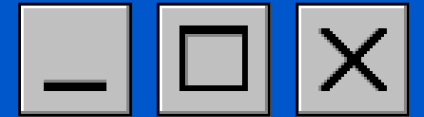
```
Recovered-D...Alabd (88))* ✨ ✕ r5glbffy..sql...6\Alabd (55))
61
62 create table passenger (
63     passengerId int primary key,
64     name varchar(100) not null,
65     passengerNo varchar(50) not null,
66     department varchar(50),
67     email varchar(100),
68     address varchar(200) not null
69 );
70
71 insert into passenger values
72 (1, 'Ahmed Ali', 'P1001', 'IT', 'ahmed@gmail.com', 'Cairo'),
73 (2, 'Mona Hassan', 'P1002', 'HR', 'mona@gmail.com', 'Giza'),
74 (3, 'Omar Salah', 'P1003', 'Finance', 'omar@gmail.com', 'Alex'),
75 (4, 'Sara Adel', 'P1004', 'Sales', 'sara@gmail.com', 'Aswan'),
76 (5, 'Youssef Kamal', 'P1005', 'Marketing', 'youssef@gmail.com', 'Luxor');
77
```



BOOKING

```
Recovered-D...Alabd (88))* ✨ ✕ r5glbffy..sql...6\Alabd (55))
79
80 create table booking (
81     bookingId int primary key,
82     paymentStatus varchar(50) not null,
83     bookingDate date not null,
84     passengerId int not null,
85     flightId int not null,
86     foreign key (passengerId) references passenger(passengerId)
87         on delete cascade
88         on update cascade,
89     foreign key (flightId) references flight(flightId)
90         on delete cascade
91         on update cascade
92 );
93
94 insert into booking values
95 (1, 'paid', '2024-12-01', 1, 1),
96 (2, 'paid', '2024-12-02', 2, 2),
97 (3, 'pending', '2024-12-03', 3, 3),
98 (4, 'paid', '2024-12-04', 4, 4),
99 (5, 'cancelled', '2024-12-05', 5, 5);
100
```





TICKET

Recovered-D...Alabd (88))* ✕ r5glbffy..sql...6\Alabd (55))

```
103 |
104 | create table ticket (
105 |     ticketNo int,
106 |     bookingId int not null,
107 |     issueDate date not null,
108 |     price decimal(10,2) not null,
109 |     seatNo varchar(10) not null,
110 |     primary key (ticketNo, bookingId),
111 |     foreign key (bookingId) references booking(bookingId)
112 |         on delete cascade
113 |         on update cascade
114 | );
115 |
116 | insert into ticket values
117 | (101,1,'2024-12-01',1500,'A1'),
118 | (102,2,'2024-12-02',1200,'B2'),
119 | (103,3,'2024-12-03',1800,'C3'),
120 | (104,4,'2024-12-04',2000,'D4'),
121 | (105,5,'2024-12-05',1600,'E5');
122 |
```

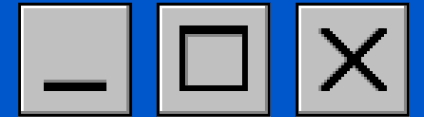


SECURITY CHECK

Recovered-D...Alabd (88))* ✕ r5glbffy..sql...6\Alabd (55))

```
124 |
125 | create table securityCheck (
126 |     checkid int primary key,
127 |     time datetime not null,
128 |     status varchar(50) not null,
129 |     passengerId int not null,
130 |     foreign key (passengerId) references passenger(passengerId)
131 |         on delete cascade
132 |         on update cascade
133 | );
134 |
135 | insert into securityCheck values
136 | (1,'2025-01-01 07:30','clear',1),
137 | (2,'2025-01-02 08:30','clear',2),
138 | (3,'2025-01-03 10:30','hold',3),
139 | (4,'2025-01-04 12:30','clear',4),
140 | (5,'2025-01-05 14:30','clear',5);
141 |
```





PASSPORT

Recovered-D...Alabd (88)* ✨ ✕ r5glbffy..sql...6\Alabd (55))

```
142 |
143 | create table passport (
144 |     passportId int primary key,
145 |     nationality varchar(50) not null,
146 |     issueDate date not null,
147 |     expiry_date date not null,
148 |     gender varchar(10) not null,
149 |     passengerId int not null,
150 | foreign key (passengerId) references passenger(passengerId)
151 |     on delete cascade
152 |     on update cascade
153 | );
154 |
155 | insert into passport values
156 | (1, 'Egyptian', '2020-01-01', '2030-01-01', 'male', 1),
157 | (2, 'Egyptian', '2019-02-01', '2029-02-01', 'female', 2),
158 | (3, 'Egyptian', '2018-03-01', '2028-03-01', 'male', 3),
159 | (4, 'Egyptian', '2021-04-01', '2031-04-01', 'female', 4),
160 | (5, 'Egyptian', '2022-05-01', '2032-05-01', 'male', 5);
```

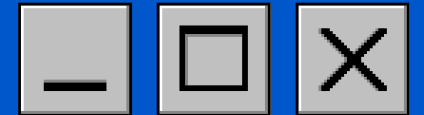


BAGGAGE

Recovered-D...Alabd (88)* ✨ ✕ r5glbffy..sql...6\Alabd (55))

```
163 |
164 | create table baggage (
165 |     bag_no int,
166 |     passengerId int not null,
167 |     status varchar(50),
168 |     weight int not null,
169 |     primary key (bag_no, passengerId),
170 | foreign key (passengerId) references passenger(passengerId)
171 |     on delete cascade
172 |     on update cascade
173 | );
174 |
175 | insert into baggage values
176 | (1, 1, 'checked', 20),
177 | (2, 2, 'checked', 18),
178 | (3, 3, 'lost', 22),
179 | (4, 4, 'checked', 25),
180 | (5, 5, 'checked', 19);
```





STAFF

Recovered-D...Alabd (88))* r5glbffy..sql...6\Alabd (55))

```
184|
185 create table staff (
186     staffId int primary key,
187     role varchar(50) not null,
188     name varchar(100) not null,
189     department varchar(50),
190     email varchar(100),
191     address varchar(200)
192 );
193
194 insert into staff values
195 (1, 'Pilot', 'Ali Mahmoud', 'Flight', 'ali@gmail.com', 'Cairo'),
196 (2, 'Co-Pilot', 'Hany Mostafa', 'Flight', 'hany@gmail.com', 'Giza'),
197 (3, 'Engineer', 'Karim Said', 'Maintenance', 'karim@gmail.com', 'Alex'),
198 (4, 'Security', 'Mohamed Adel', 'Security', 'mohamed@gmail.com', 'Aswan'),
199 (5, 'Hostess', 'Nour Hassan', 'Cabin', 'nour@gmail.com', 'Luxor');
```

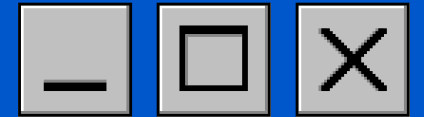


WORKS_ON

Recovered-D...Alabd (88))* r5glbffy..sql...6\Alabd (55))

```
205 create table works_on (
206     staffId int not null,
207     flightId int not null,
208     primary key (staffId, flightId),
209     foreign key (staffId) references staff(staffId)
210         on delete cascade
211         on update cascade,
212     foreign key (flightId) references flight(flightId)
213         on delete cascade
214         on update cascade
215 );
216
217 insert into works_on values
218 (1,1),
219 (2,2),
220 (3,3),
221 (4,4),
222 (5,5);
```





PASSENGER_ADDRESS

Recovered-D...Alabd (88))* ✨ ✕ r5glbffy..sql...6\Alabd (55))

```
225 |
226 create table passenger_address (
227     passengerId int not null,
228     address varchar(200) not null,
229     primary key (passengerId, address),
230     foreign key (passengerId) references passenger(passengerId)
231         on delete cascade
232         on update cascade
233 );
234
235 insert into passenger_address values
236 (1, 'Cairo'),
237 (2, 'Giza'),
238 (3, 'Alexandria'),
239 (4, 'Aswan'),
240 (5, 'Luxor');
```



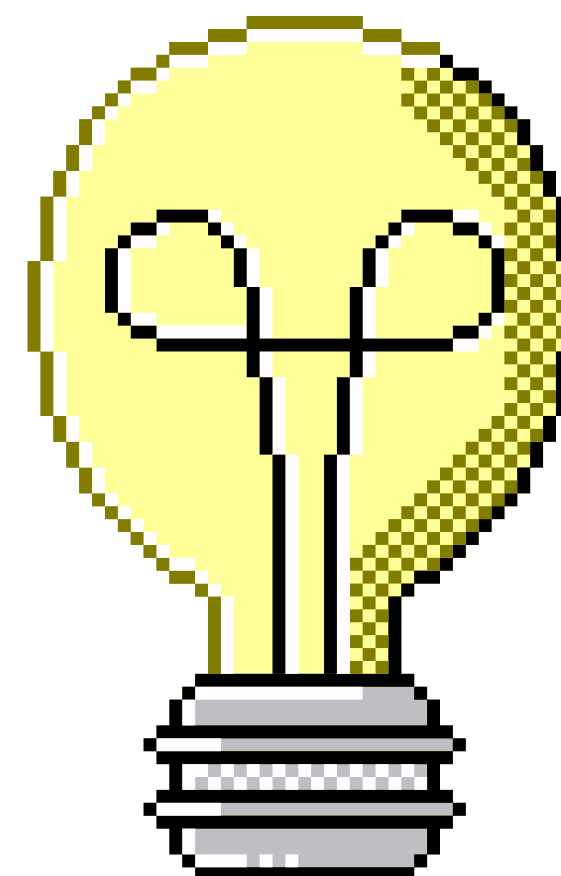
STAFF_ADDRESS

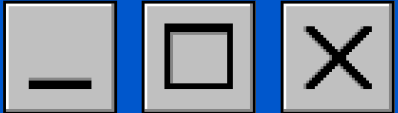
```
241
242 create table staff_address (
243     staffId int not null,
244     address varchar(200) not null,
245     primary key (staffId, address),
246     foreign key (staffId) references staff(staffId)
247         on delete cascade
248         on update cascade
249 );
250
251 insert into staff_address values
252 (1, 'Cairo'),
253 (2, 'Giza'),
254 (3, 'Alexandria'),
255 (4, 'Aswan'),
256 (5, 'Luxor');
```





JOIN STATEMENT





INNER & LEFT JOIN

```
263 -----
264
265 -- INNER JOIN
266 -----
267 select flight.flightNo , airport.name as airport_name
268 from flight
269 inner join airport
270 on flight.airportId = airport.airportId;
271
272 -----
```

32 % × 55 ⚠ 0 ↑ ↓

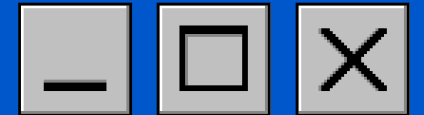
Results		Messages
	flightNo	airport_name
1	MS101	Cairo Airport
2	MS102	Alex Airport
3	QR201	Aswan Airport
4	EK301	Luxor Airport
5	SV401	Hurghada Airport

```
274 -----
275 select passenger.name , booking.paymentStatus
276 from passenger
277 left join booking
278 on passenger.passengerId = booking.passengerId;
279
280 -----
```

82 % × 55 ⚠ 0 ↑ ↓

Results		Messages
	name	paymentStatus
1	Ahmed Ali	paid
2	Mona Hassan	paid
3	Omar Salah	pending
4	Sara Adel	paid
5	Youssef Kamal	cancelled





RIGHT & FULL
OUTER JOIN

```
283  
284 select booking.bookingId , passenger.name  
285 from booking  
286 right join passenger  
287 on booking.passengerId = passenger.passengerId;  
288  
289
```

82 % 55 0

Results Messages

	bookingId	name
1	1	Ahmed Ali
2	2	Mona Hassan
3	3	Omar Salah
4	4	Sara Adel
5	5	Youssef Kamal

```
295 select passenger.name , booking.bookingId  
296 from passenger  
297 full outer join booking  
298 on passenger.passengerId = booking.passengerId;
```

82 % 55 0

Results Messages

	name	bookingId
1	Ahmed Ali	1
2	Mona Hassan	2
3	Omar Salah	3
4	Sara Adel	4
5	Youssef Kamal	5



CROSS & SELF JOIN

```
305 select airport.name , gate.gateNumber
306 from airport
307 cross join gate;
```

82 %



55



0



Results Messages

	name	gateNumber
1	Cairo Airport	G1
2	Alex Airport	G1
3	Aswan Airport	G1
4	Luxor Airport	G1
5	Hurghada Airport	G1
6	Cairo Airport	G2
7	Alex Airport	G2
8	Aswan Airport	G2
9	Luxor Airport	G2

```
314 select s1.name as staff1 , s2.name as staff2 , s1.department
315 from staff s1
316 join staff s2
317 on s1.department = s2.department
318 and s1.staffId <> s2.staffId;
319
```

82 %



55

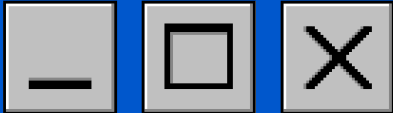


0



Results Messages

	staff1	staff2	department
1	Ali Mahmoud	Hany Mostafa	Flight
2	Hany Mostafa	Ali Mahmoud	Flight



RIGHT & LEFT
JOIN

```
324 select flight.flightNo , gate.gateNumber
325 from flight
326 left join gate
327 on flight.gateId = gate.gateId;
328
```

82 % ✖ 55 ⚠ 0 ↑ ↓

Results Messages

	flightNo	gateNumber
1	MS101	G1
2	MS102	G2
3	QR201	G3
4	EK301	G4
5	SV401	G5

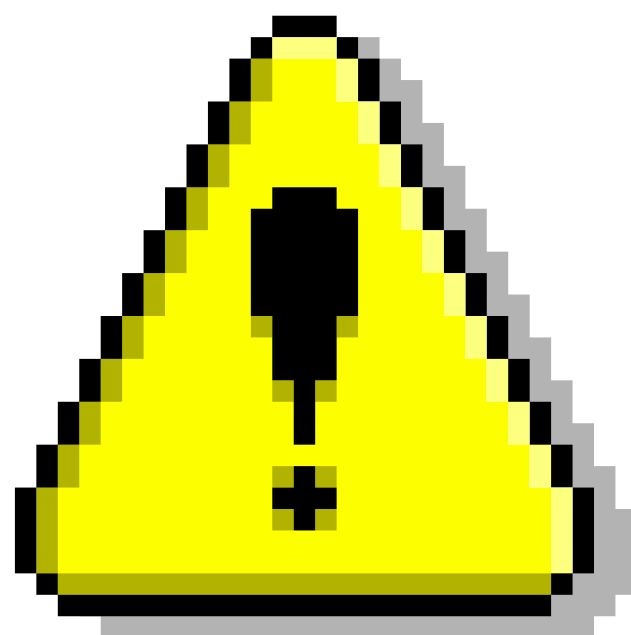
```
333 select staff.name , works_on.flightId
334 from works_on
335 right join staff
336 on works_on.staffId = staff.staffId;
337
```

82 % ✖ 55 ⚠ 0 ↑ ↓

Results Messages

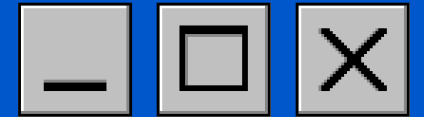
	name	flightId
1	Ali Mahmoud	1
2	Hany Mostafa	2
3	Karim Said	3
4	Mohamed Adel	4
5	Nour Hassan	5





PROCEDURE





Procedure 1

```
344 -----
345 -- PROCEDURE 1
346 -----
347 create procedure getPassengerBookings
348 as
349 begin
350     select passenger.name ,
351           booking.bookingId ,
352           booking.paymentStatus
353 from passenger
354 inner join booking
355 on passenger.passengerId = booking.passengerId;]
356 end;
357
358 -----
```

82 %

✓ No issues found

Messages

Commands completed successfully.

Completion time: 2026-05-09T15:56:01.7950962+03:00

Procedure 2

```
363 -----
364 -- PROCEDURE 2
365 -----
366 create procedure getFlightDetails
367 as
368 begin
369     select flight.flightNo ,
370           airport.name as airport_name ,
371           gate.gateNumber ,
372           flight.status
373 from flight
374 inner join airport
375 on flight.airportId = airport.airportId
376 inner join gate
377 on flight.gateId = gate.gateId;
378 end;
379
```

82 %

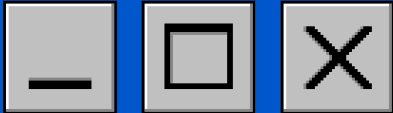
✓ No issues found

Messages

Commands completed successfully.

Completion time: 2026-05-09T15:56:01.7950962+03:00





379
380
381
382
383
384
385

```
-- EXECUTE  
exec getFlightDetails;
```

82 % × 55 ⚠ 0 ↑ ↓

Results Messages

	flightNo	airport_name	gateNumber	status
1	MS101	Cairo Airport	G1	on time
2	MS102	Alex Airport	G2	delayed
3	QR201	Aswan Airport	G3	on time
4	EK301	Luxor Airport	G4	cancelled
5	SV401	Hurghada Airport	G5	on time



EXECUTE

Cancel

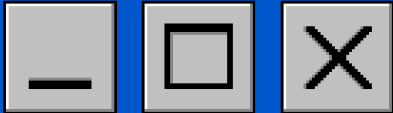
Yes





FUNCTION





D

```
388 -----
389 -- FUNCTION 1
390 -----
391 create function totalTickets()
392 returns int
393 as
394 begin
395     declare @total int;
396
397     select @total = count(*)
398     from ticket;
399
400     return @total;
401 end;
402
403 -----
404 -- EXECUTE
405 -----
406 select dbo.totalTickets() as total_tickets;
```

82 % v x 55 ! 0 ↑ ↓

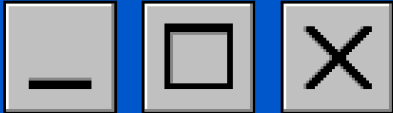
Results Messages

	total_tickets
1	5

 FUNCTION

Cancel

Yes



```
408 -----
409 -- FUNCTION 2
410 -----
411 create function passengerCountByCity(@city varchar(100))
412 returns int
413 as
414 begin
415     declare @count int;
416
417     select @count = count(*)
418     from passenger
419     where address = @city;
420
421     return @count;
422 end;
423
424 -----
425 -- EXECUTE
426 -----
427 select dbo.passengerCountByCity('Cairo') as passengers_in_cairo;
```

82 % × 55 ⚠ 0 ↑ ↓

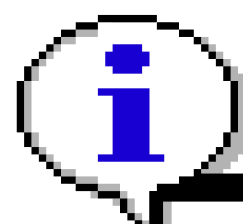
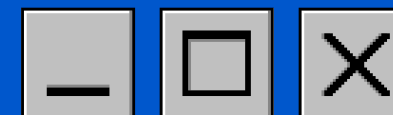
Results Messages

	passengers_in_cairo
1	1

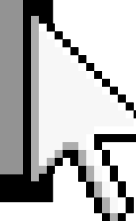
FUNCTION

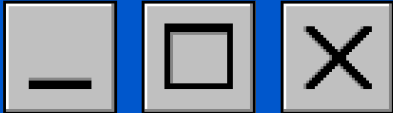
Cancel

Yes



TRIGGERS





```
435 -----
436 -- TRIGGER 1
437 -----
438 create trigger trg_after_insert_booking
439 on booking
440 after insert
441 as
442 begin
443     print 'New booking inserted successfully';
444 end;
445 -----
446 -- TEST TRIGGER
447 -----
448
449 insert into booking values
450 (6, 'paid', '2024-12-06', 1, 2);
451 -----
```

82 %

✖

55

⚠

0

↑

↓

Messages

New booking inserted successfully

(1 row affected)

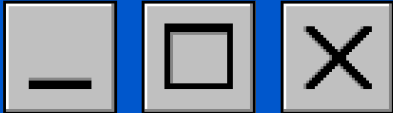
Completion time: 2026-05-09T18:53:12.4519151+03:00

i

INSERT

Cancel

Yes



```
452 -----
453 -- TRIGGER 2
454 -----
455 create trigger trg_after_delete_passenger
456 on passenger
457 after delete
458 as
459 begin
460     print 'Passenger deleted successfully';
461 end;
462 -----
463 -- TEST TRIGGER
464 -----
465
466 delete from passenger
467 where passengerId = 5;
468
469
```

82 % × 55 ⚠ 0 ↑ ↓

Messages

Passenger deleted successfully

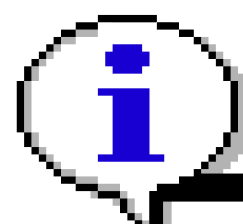
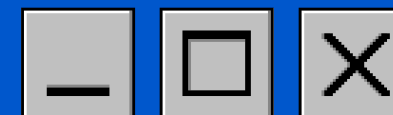
(1 row affected)

Completion time: 2026-05-09T18:55:41.7716970+03:00

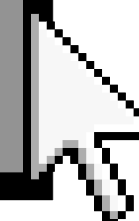
 DELETE

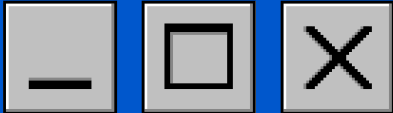
Cancel

Yes



VIEW





```
475 -----
476 -- VIEW 1
477 -----
478 create view delayed_flights
479 as
480 select *
481 from flight
482 where status = 'delayed';
483 -----
484
485 -- VIEW 2
486 -----
487 create view staff_emails
488 as
489 select name , email
490 from staff;
491
492
493
```

82 %

✖

55

⚠

0

↑

↓

📄 Messages

Commands completed successfully.

Completion time: 2026-05-09T18:56:41.8117425+03:00

i

VIEW

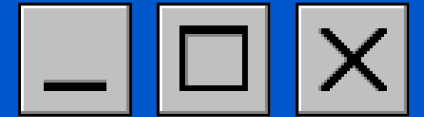
Cancel

Yes



RELATIONAL ALGEBRA OPERATIONS





RUN

```
499 -----
500 -- 1. SELECTION ( $\sigma$ )
501 -----
502 select *
503 from passenger
504 where address = 'Cairo';
505 -----
```

82 % 55 0

Results Messages

	passengerId	name	passengerNo	department	email	address
1	1	Ahmed Ali	P1001	IT	ahmed@gmail.com	Cairo

```
506 -----
507 -- 2. PROJECTION ( $\pi$ )
508 -----
509 select name , email
510 from passenger;
511 -----
```

82 % 55 0

Results Messages

	name	email
1	Ahmed Ali	ahmed@gmail.com
2	Mona Hassan	mona@gmail.com
3	Omar Salah	omar@gmail.com
4	Sara Adel	sara@gmail.com





RUN

```
511  
512  
513 -- 3. RENAME (p)  
514  
515 select name as passenger_name ,  
516        address as city  
517 from passenger;
```

82 %



55



0



Results Messages

	passenger_name	city
1	Ahmed Ali	Cairo
2	Mona Hassan	Giza
3	Omar Salah	Alex
4	Sara Adel	Aswan

```
519  
520 -- 4. DIFFERENCE (-)  
521  
522 select passengerId  
523 from passenger  
524  
525 except  
526  
527 select passengerId  
528 from booking;  
529
```

82 %



55



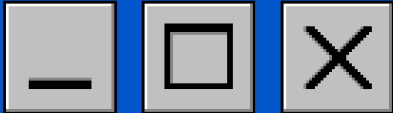
0



Results Messages

passengerId





RUN

```
530
531
532 -- 5. UNION (U)
533
534 select address
535 from passenger
536
537 union
538
539 select address
540 from staff;
541
```

82 % 55 0

Results Messages

	address
1	Alex
2	Aswan
3	Cairo
4	Giza
5	Luxor

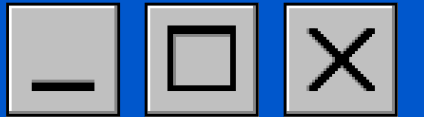
```
542
543 -- 6. CARTESIAN PRODUCT (x)
544
545 select *
546 from airport
547 cross join gate;
```

82 % 55 0

Results Messages

	airportId	name	location	terminalsCount	gateId	gateNumber	terminal
1	1	Cairo Airport	Cairo	3	1	G1	T1
2	2	Alex Airport	Alexandria	2	1	G1	T1
3	3	Aswan Airport	Aswan	1	1	G1	T1
4	4	Luxor Airport	Luxor	2	1	G1	T1
5	5	Hurghada Airport	Hurghada	3	1	G1	T1
6	1	Cairo Airport	Cairo	3	2	G2	T1
7	2	Alex Airport	Alexandria	2	2	G2	T1
8	3	Aswan Airport	Aswan	1	2	G2	T1
9	4	Luxor Airport	Luxor	2	2	G2	T1





THIS IS THE
END

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